

1. Development Phase: Description

The development phase is the stage where developers write code to transform design documents and architectural plans into a functional software product.

2. Front-End vs. Back-End

- **Front-End:** Focuses on the client-side interface, including layout, user inputs, and visual design.
- **Back-End:** Focuses on the server-side, handling data processing, database management, and API logic.

3. Common Tools

1. **Git/GitHub:** Used for version control and collaborative code management.
2. **Visual Studio Code:** A code editor used for writing, testing, and debugging source code.
3. **Jira:** Used for tracking tasks, bugs, and overall development progress.

4. Importance of Version Control

Version control provides a "source of truth," enabling teams to track history, revert to previous stable states, and collaborate on the same codebase without overwriting work.

5. Case Study: Knight Capital Group (2012)

Knight Capital lost \$440 million when a deployment error left legacy code running on a production server, causing it to execute erroneous, high-speed trades.

References:

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2. Coursera. (2026). *What Is Version Control? How It Works*.
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3. PRMIA. (2026). *The Knight Capital Algorithmic Trading Failure*.
<https://prmia.org/common/Uploaded%20files/eAI/PRMIA%20Case%20study%20-%20Knight%20Trading.pdf>